

ADVANCED COMPUTER NETWORKS
(MTITE101)

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Practical 6:

1. Aim

To implement and compare Distance Vector (RIPv2) and Link State (Global SPF) routing protocols in ns-3 across a diamond network topology, evaluating convergence time, route computation, and protocol overhead.

2. Step-by-Step Implementation Guide

Step 1: Environment & Directory Setup

Navigate to your ns-3 root folder:

Bash
```
cd ~/ns-allinone-3.33/ns-3.33
```

Step 2: Script Creation

Create a new file named `dv_vs_ls.py` inside the `scratch/` directory.

Step 3: Script Execution

Execute for Distance Vector mode:

Bash
```
./waf --pyrun scratch/dv_vs_ls.py
```

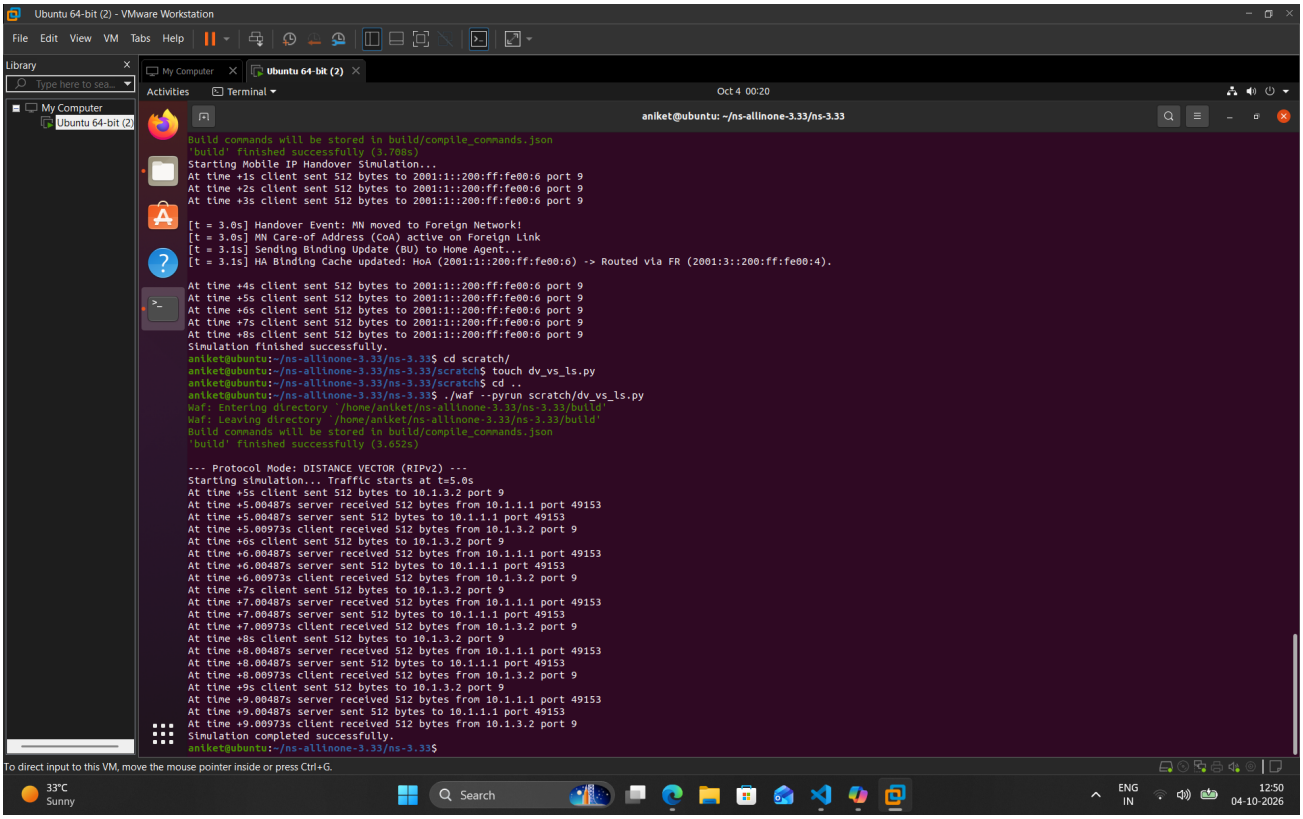
To run Link State mode, set `USE_DISTANCE_VECTOR = False` inside `scratch/dv_vs_ls.py` and execute again.

3. Protocol Architecture Comparison

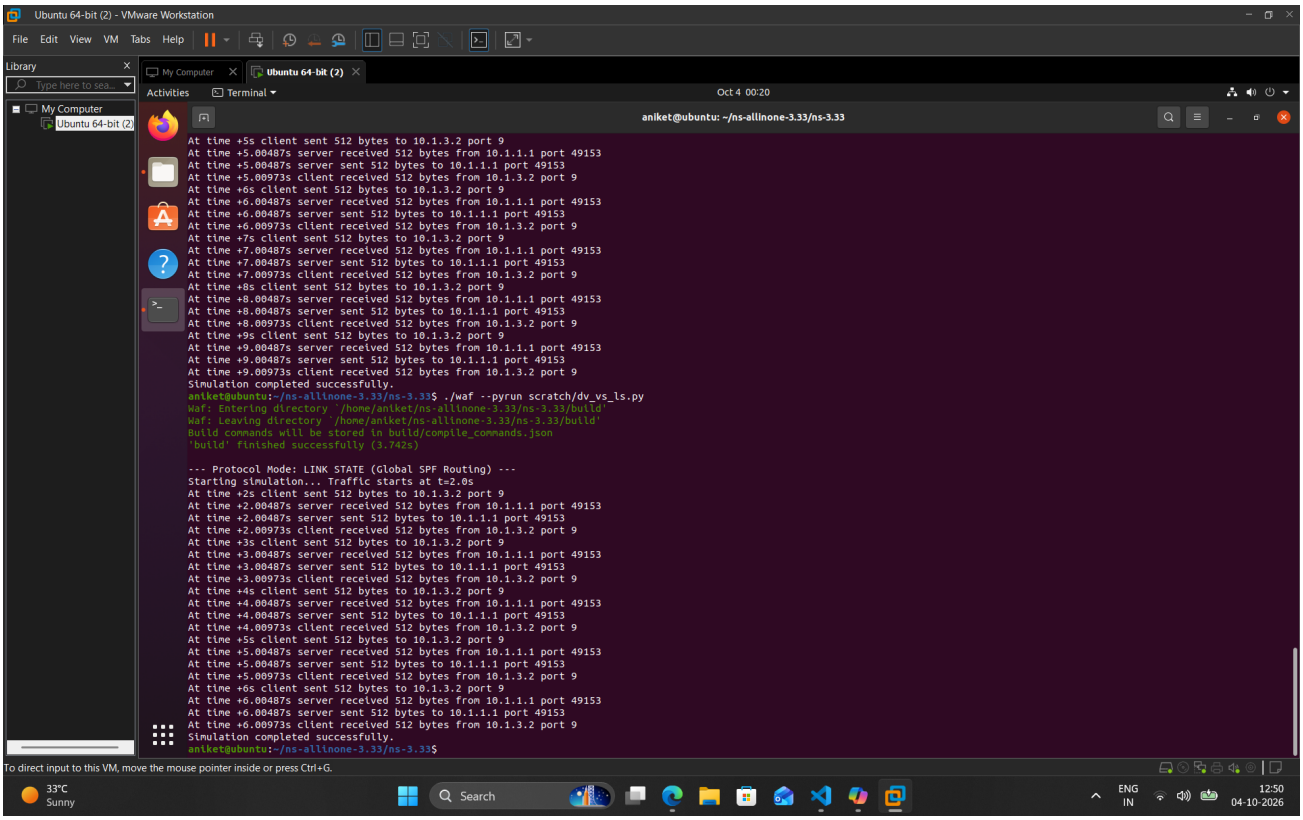
- **Distance Vector (`ns.internet.RipHelper`)**: Implements the Bellman-Ford algorithm. Nodes periodically exchange routing tables with immediate neighbors. Convergence requires multiple exchange rounds across intermediate hops.
- **Link State (`ns.internet.Ipv4GlobalRoutingHelper`)**: Implements Dijkstra's Shortest Path First (SPF) algorithm. Every router constructs a global graph of the entire network topology and pre-computes optimal paths before traffic flows begin.

4. Execution Output & Terminal Logs

Mode A: Distance Vector (RIPv2)



Mode B: Link State (Global Routing)



5. Comparative Analysis

Feature / Metric	Distance Vector (RIPv2)	Link State (Global Routing)
Convergence Time	Delayed (delay for full convergence)	Immediate (pre-computed database)
Routing Information	Neighbor distance vectors ("Routing by rumor")	Complete topology map on every node
Traffic Start Window	Traffic delayed to to prevent packet drops	Traffic scheduled as early as
Control Overhead	High periodic bandwidth consumption	High CPU/Memory utilization during SPF calculation